5th Week

데이터 로딩과 저장, 파일 형식

강의 내용

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1부: 데이터 로딩과 저장, 파일형식

1) 텍스트 파일에서 데이터를 읽고 쓰는 법

2) 이진 데이터 형식

3) 웹 API와 함께 사용하기

4) 데이터베이스와 함께 사용하기

데이터 다루기

Reading and Writing Data in Text Format

- 데이터를 다루는 것은 수업 때 배울 대부분의 도구를 사용하는 데 필요한 첫 번째 단계
- 다양한 형식의 데이터를 읽고 쓰는 데 도움이 되는 수많은 도구가 있지만, pandas를 사용한 데이터 입력 및 출력을 진행할 예정!
- 입력 및 출력 구분
  - 텍스트 파일
  - 보다 효율적인 디스크 형식 읽기
  - 데이터베이스에서 데이터 로드
  - 웹 API와 같은 네트워크 소스와의 상호 작용

텍스트 파일에서 데이터를 읽고 쓰는 법

Reading and Writing Data in Text Format

pandas features a number of functions for reading tabular data as a DataFrame object. Table 6-1 summarizes some of them, though `read_csv` and `read_table` are likely the ones you'll use the most.

Table 6-1. Parsing functions in pandas

Function	Description
`read_csv`	Load delimited data from a file, URL, or file-like object; use comma as default delimiter
`read_table`	Load delimited data from a file, URL, or file-like object; use tab (`'\t'`) as default delimiter
`read_fwf`	Read data in fixed-width column format (i.e., no delimiters)
`read_clipboard`	Version of `read_table` that reads data from the clipboard; useful for converting tables from web pages
`read_excel`	Read tabular data from an Excel XLS or XLSX file
`read_hdf`	Read HDF5 files written by pandas
`read_html`	Read all tables found in the given HTML document
`read_json`	Read data from a JSON (JavaScript Object Notation) string representation
`read_msgpack`	Read pandas data encoded using the MessagePack binary format
`read_pickle`	Read an arbitrary object stored in Python pickle format
`read_sas`	Read a SAS dataset stored in one of the SAS system's custom storage formats
`read_sql`	Read the results of a SQL query (using SQLAlchemy) as a pandas DataFrame
`read_stata`	Read a dataset from Stata file format
`read_feather`	Read the Feather binary file format

I'll give an overview of the mechanics of these functions, which are meant to convert text data into a DataFrame. The optional arguments for these functions may fall into a few categories:

Indexing
: Can treat one or more columns as the returned DataFrame, and whether to get column names from the file, the user, or not at all.

Type inference and data conversion
: This includes the user-defined value conversions and custom list of missing value markers.

Datetime parsing
: Includes combining capability, including combining date and time information spread over multiple columns into a single column in the result.

Iterating
: Support for iterating over chunks of very large files.

Unclean data issues
: Skipping rows or a footer, comments, or other minor things like numeric data with thousands separated by commas.

텍스트 파일에서 데이터를 읽고 쓰는 법

Reading and Writing Data in Text Format

- 현실 세계의 데이터는 매우 지저분할 수 있기 때문에,
  일부 데이터 로드 기능(특히 read_csv)은 시간이 지남에 따라 옵션이 매우 복잡해졌다
- 따라서, 다양한 매개변수의 수에 압도당하는 것은 정상이다 (read_csv의 parameter는 50개가 넘는다)
- Pandas Documentation에는 각각의 작동 방식에 대한 많은 예가 있으므로 특정 파일을 읽는 데 어려움을 겪고 있다면 적절한 매개변수를 찾는 데 도움이 되는 유사한 예를 충분히 찾아볼 수 있다
- **pandas.read_csv와 같은 함수** 중 일부는 **열 데이터 타입**이 데이터 포맷의 일부가 아니기 때문에 **열 변수의 타입 유추를 추가 수행**한다
  - 즉, 어떤 열이 숫자, 정수, 부울 또는 문자열인지 지정할 필요가 없다.
  - 또한 HDF5, Feather 및 msgpack과 같은 다른 데이터 포맷에는 열 데이터 타입이 포맷에 저장되어 있다

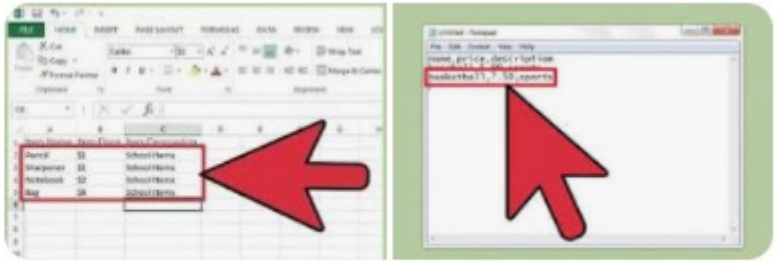
CSV(영어: comma-separated values)는 몇 가지 필드를 쉼표(,)로 구분한 텍스트 데이터 및 텍스트 파일이다. 확장자는 .**csv**이며 MIME 형식은 text/**csv**이다. comma-separated variables라고도 한다.

표준: RFC 4180

포맷 종류: 텍스트

https://ko.wikipedia.org › wiki › CSV_(파일_형식)

CSV (파일 형식) - 위키백과, 우리 모두의 백과사전



텍스트 파일에서 데이터를 읽고 쓰는 법

Reading and Writing Data in Text Format

Handling dates and other custom types can require extra effort. Let's start with a small comma-separated (CSV) text file:

```
In [8]: !cat examples/ex1.csv
a,b,c,d,message
1,2,3,4,hello
5,6,7,8,world
9,10,11,12,foo
```

Here I used the Unix `cat` shell command to print the raw contents of the file to the screen. If you're on Windows, you can use `type` instead of `cat` to achieve the same effect.

Since this is comma-delimited, we can use `read_csv` to read it into a DataFrame:

```
In [9]: df = pd.read_csv('examples/ex1.csv')

In [10]: df
Out[10]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

We could also have used `read_table` and specified the delimiter:

```
In [11]: pd.read_table('examples/ex1.csv', sep=',')
Out[11]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

A file will not always have a header row. Consider this file:

```
In [12]: !cat examples/ex2.csv
1,2,3,4,hello
5,6,7,8,world
9,10,11,12,foo
```

To read this file, you have a couple of options. You can allow pandas to assign default column names, or you can specify names yourself:

```
In [13]: pd.read_csv('examples/ex2.csv', header=None)
Out[13]:
   0   1   2   3      4
0  1   2   3   4  hello
1  5   6   7   8  world
2  9  10  11  12    foo

In [14]: pd.read_csv('examples/ex2.csv', names=['a', 'b', 'c', 'd', 'message'])
Out[14]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

Suppose you wanted the `message` column to be the index of the returned DataFrame. You can either indicate you want the column at index 4 or named `'message'` using the `index_col` argument:

```
In [15]: names = ['a', 'b', 'c', 'd', 'message']

In [16]: pd.read_csv('examples/ex2.csv', names=names, index_col='message')
Out[16]:
         a   b   c   d
message
hello    1   2   3   4
world    5   6   7   8
foo      9  10  11  12
```

텍스트 파일에서 데이터를 읽고 쓰는 법

Reading and Writing Data in Text Format

In the event that you want to form a hierarchical index from multiple columns, pass a list of column numbers or names:

```
In [17]: !cat examples/csv_mindex.csv
key1,key2,value1,value2
one,a,1,2
one,b,3,4
one,c,5,6
one,d,7,8
two,a,9,10
two,b,11,12
two,c,13,14
two,d,15,16

In [18]: parsed = pd.read_csv('examples/csv_mindex.csv',
   ....:                      index_col=['key1', 'key2'])

In [19]: parsed
Out[19]:
           value1  value2
key1 key2
one  a          1       2
     b          3       4
     c          5       6
     d          7       8
two  a          9      10
     b         11      12
     c         13      14
     d         15      16
```

In some cases, a table might not have a fixed delimiter, using whitespace or some other pattern to separate fields. Consider a text file that looks like this:

```
In [20]: list(open('examples/ex3.txt'))
Out[20]:
['            A         B         C\n',
 'aaa -0.264438 -1.026059 -0.619500\n',
 'bbb  0.927272  0.302904 -0.032399\n',
 'ccc -0.264273 -0.386314 -0.217601\n',
 'ddd -0.871858 -0.348382  1.100491\n']
```

While you could do some munging by hand, the fields here are separated by a variable amount of whitespace. In these cases, you can pass a regular expression as a delimiter for `read_table`. This can be expressed by the regular expression `\s+`, so we have then:

```
In [21]: result = pd.read_table('examples/ex3.txt', sep='\s+')

In [22]: result
Out[22]:
            A         B         C
aaa -0.264438 -1.026059 -0.619500
bbb  0.927272  0.302904 -0.032399
ccc -0.264273 -0.386314 -0.217601
ddd -0.871858 -0.348382  1.100491
```

Because there was one fewer column name than the number of data rows, `read_table` infers that the first column should be the DataFrame's index in this special case.

텍스트 파일에서 데이터를 읽고 쓰는 법

Reading and Writing Data in Text Format

The parser functions have many additional arguments to help you handle the wide variety of exception file formats that occur (see a partial listing in Table 6-2). For example, you can skip the first, third, and fourth rows of a file with `skiprows`:

```
In [23]: !cat examples/ex4.csv
# hey!
a,b,c,d,message
# just wanted to make things more difficult for you
# who reads CSV files with computers, anyway?
1,2,3,4,hello
5,6,7,8,world
9,10,11,12,foo
In [24]: pd.read_csv('examples/ex4.csv', skiprows=[0, 2, 3])
Out[24]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

Handling missing values is an important and frequently nuanced part of the file parsing process. Missing data is usually either not present (empty string) or marked by some *sentinel* value. By default, pandas uses a set of commonly occurring sentinels, such as `NA` and `NULL`:

```
In [25]: !cat examples/ex5.csv
something,a,b,c,d,message
one,1,2,3,4,NA
two,5,6,,8,world
three,9,10,11,12,foo
In [26]: result = pd.read_csv('examples/ex5.csv')
In [27]: result
Out[27]:
  something  a   b     c   d message
0       one  1   2   3.0   4     NaN
1       two  5   6   NaN   8   world
2     three  9  10  11.0  12     foo

In [28]: pd.isnull(result)
Out[28]:
   something      a      b      c      d  message
0      False  False  False  False  False     True
1      False  False  False   True  False    False
2      False  False  False  False  False    False
```

텍스트 파일에서 데이터를 읽고 쓰는 법

Reading and Writing Data in Text Format

The na_values option can take either a list or set of strings to consider missing values:

```
In [29]: result = pd.read_csv('examples/ex5.csv', na_values=['NULL'])

In [30]: result
Out[30]:
  something  a   b     c   d message
0       one  1   2   3.0   4     NaN
1       two  5   6   NaN   8   world
2     three  9  10  11.0  12     foo
```

Different NA sentinels can be specified for each column in a dict:

```
In [31]: sentinels = {'message': ['foo', 'NA'], 'something': ['two']}

In [32]: pd.read_csv('examples/ex5.csv', na_values=sentinels)
Out[32]:
  something  a   b     c   d message
0       one  1   2   3.0   4     NaN
1       NaN  5   6   NaN   8   world
2     three  9  10  11.0  12     NaN
```

Table 6-2. Some read_csv/read_table function arguments

Argument	Description
path	String indicating filesystem location, URL, or file-like object
sep or delimiter	Character sequence or regular expression to use to split fields in each row
header	Row number to use as column names; defaults to 0 (first row), but should be None if there is no header row
index_col	Column numbers or names to use as the row index in the result; can be a single name/number or a list of them for a hierarchical index
names	List of column names for result, combine with header=None
skiprows	Number of rows at beginning of file to ignore or list of row numbers (starting from 0) to skip.
na_values	Sequence of values to replace with NA.
comment	Character(s) to split comments off the end of lines.
parse_dates	Attempt to parse data to datetime; False by default. If True, will attempt to parse all columns. Otherwise can specify a list of column numbers or name to parse. If element of list is tuple or list, will combine multiple columns together and parse to date (e.g., if date/time split across two columns).
keep_date_col	If joining columns to parse date, keep the joined columns; False by default.
converters	Dict containing column number of name mapping to functions (e.g., {'foo': f} would apply the function f to all values in the 'foo' column).
dayfirst	When parsing potentially ambiguous dates, treat as international format (e.g., 7/6/2012 -> June 7, 2012); False by default.
date_parser	Function to use to parse dates.
nrows	Number of rows to read from beginning of file.
iterator	Return a TextParser object for reading file piecemeal.
chunksize	For iteration, size of file chunks.
skip_footer	Number of lines to ignore at end of file.
verbose	Print various parser output information, like the number of missing values placed in non-numeric columns.
encoding	Text encoding for Unicode (e.g., 'utf-8' for UTF-8 encoded text).
squeeze	If the parsed data only contains one column, return a Series.
thousands	Separator for thousands (e.g., ',' or '.').

텍스트 파일에서 데이터를 읽고 쓰는 법 > 텍스트 파일 조금씩 읽어오기

When processing very large files or figuring out the right set of arguments to correctly process a large file, you may only want to read in a small piece of a file or iterate through smaller chunks of the file.

Before we look at a large file, we make the pandas display settings more compact:

```
In [33]: pd.options.display.max_rows = 10
```

Now we have:

```
In [34]: result = pd.read_csv('examples/ex6.csv')

In [35]: result
Out[35]:
           one       two     three      four key
0     0.467976 -0.038649 -0.295344 -1.824726   L
1    -0.358893  1.404453  0.704965 -0.200638   B
2    -0.501840  0.659254 -0.421691 -0.057688   G
3     0.204886  1.074134  1.388361 -0.982404   R
4     0.354628 -0.133116  0.283763 -0.837063   Q
...        ...       ...       ...       ...  ..
9995  2.311896 -0.417070 -1.409599 -0.515821   L
9996 -0.479893 -0.650419  0.745152 -0.646038   E
9997  0.523331  0.787112  0.486066  1.093156   K
9998 -0.362559  0.598894 -1.843201  0.887292   G
9999 -0.096376 -1.012999 -0.657431 -0.573315   0
[10000 rows x 5 columns]
```

If you want to only read a small number of rows (avoiding reading the entire file), specify that with `nrows`:

```
In [36]: pd.read_csv('examples/ex6.csv', nrows=5)
Out[36]:
        one       two     three      four key
0  0.467976 -0.038649 -0.295344 -1.824726   L
1 -0.358893  1.404453  0.704965 -0.200638   B
2 -0.501840  0.659254 -0.421691 -0.057688   G
3  0.204886  1.074134  1.388361 -0.982404   R
4  0.354628 -0.133116  0.283763 -0.837063   Q
```

텍스트 파일에서 데이터를 읽고 쓰는 법 > 텍스트 파일 조금씩 읽어오기

To read a file in pieces, specify a chunksize as a number of rows:

```
In [37]: chunker = pd.read_csv('examples/ex6.csv', chunksize=1000)

In [38]: chunker
Out[38]: <pandas.io.parsers.TextFileReader at 0x7f6b1e2672e8>
```

The TextParser object returned by read_csv allows you to iterate over the parts of the file according to the chunksize. For example, we can iterate over ex6.csv, aggregating the value counts in the 'key' column like so:

```
chunker = pd.read_csv('examples/ex6.csv', chunksize=1000)

tot = pd.Series([])
for piece in chunker:
    tot = tot.add(piece['key'].value_counts(), fill_value=0)

tot = tot.sort_values(ascending=False)
```

We have then:

```
In [40]: tot[:10]
Out[40]:
E    368.0
X    364.0
L    346.0
O    343.0
Q    340.0
M    338.0
J    337.0
F    335.0
K    334.0
H    330.0
dtype: float64
```

TextParser is also equipped with a get_chunk method that enables you to read pieces of an arbitrary size.

텍스트 파일에서 데이터를 읽고 쓰는 법 > 데이터를 텍스트 형식으로 기록하기

Data can also be exported to a delimited format. Let's consider one of the CSV files read before:

```
In [41]: data = pd.read_csv('examples/ex5.csv')

In [42]: data
Out[42]:
  something  a   b     c   d message
0       one  1   2   3.0   4     NaN
1       two  5   6   NaN   8   world
2     three  9  10  11.0  12     foo
```

Using DataFrame's to_csv method, we can write the data out to a comma-separated file:

```
In [43]: data.to_csv('examples/out.csv')

In [44]: !cat examples/out.csv
,something,a,b,c,d,message
0,one,1,2,3.0,4,
1,two,5,6,,8,world
2,three,9,10,11.0,12,foo
```

Other delimiters can be used, of course (writing to sys.stdout so it prints the text result to the console):

```
In [45]: import sys

In [46]: data.to_csv(sys.stdout, sep='|')
|something|a|b|c|d|message
0|one|1|2|3.0|4|
1|two|5|6||8|world
2|three|9|10|11.0|12|foo
```

Missing values appear as empty strings in the output. You might want to denote them by some other sentinel value:

```
In [47]: data.to_csv(sys.stdout, na_rep='NULL')
,something,a,b,c,d,message
0,one,1,2,3.0,4,NULL
1,two,5,6,NULL,8,world
2,three,9,10,11.0,12,foo
```

With no other options specified, both the row and column labels are written. Both of these can be disabled:

```
In [48]: data.to_csv(sys.stdout, index=False, header=False)
one,1,2,3.0,4,
two,5,6,,8,world
three,9,10,11.0,12,foo
```

You can also write only a subset of the columns, and in an order of your choosing:

```
In [49]: data.to_csv(sys.stdout, index=False, columns=['a', 'b', 'c'])
a,b,c
1,2,3.0
5,6,
9,10,11.0
```

Series also has a `to_csv` method:

```
In [50]: dates = pd.date_range('1/1/2000', periods=7)

In [51]: ts = pd.Series(np.arange(7), index=dates)

In [52]: ts.to_csv('examples/tseries.csv')

In [53]: !cat examples/tseries.csv
2000-01-01,0
2000-01-02,1
2000-01-03,2
2000-01-04,3
2000-01-05,4
2000-01-06,5
2000-01-07,6
```

텍스트 파일에서 데이터를 읽고 쓰는 법 > 구분자 형식 다루기

It's possible to load most forms of tabular data from disk using functions like `pandas.read_table`. In some cases, however, some manual processing may be necessary. It's not uncommon to receive a file with one or more malformed lines that trip up `read_table`. To illustrate the basic tools, consider a small CSV file:

```
In [54]: !cat examples/ex7.csv
"a","b","c"
"1","2","3"
"1","2","3"
```

For any file with a single-character delimiter, you can use Python's built-in `csv` module. To use it, pass any open file or file-like object to `csv.reader`:

```
import csv
f = open('examples/ex7.csv')

reader = csv.reader(f)
```

Iterating through the reader like a file yields tuples of values with any quote characters removed:

```
In [56]: for line in reader:
   ....:     print(line)
['a', 'b', 'c']
['1', '2', '3']
['1', '2', '3']
```

From there, it's up to you to do the wrangling necessary to put the data in the form that you need it. Let's take this step by step. First, we read the file into a list of lines:

```
In [57]: with open('examples/ex7.csv') as f:
   ....:     lines = list(csv.reader(f))
```

Then, we split the lines into the header line and the data lines:

```
In [58]: header, values = lines[0], lines[1:]
```

Then we can create a dictionary of data columns using a dictionary comprehension and the expression `zip(*values)`, which transposes rows to columns:

```
In [59]: data_dict = {h: v for h, v in zip(header, zip(*values))}

In [60]: data_dict
Out[60]: {'a': ('1', '1'), 'b': ('2', '2'), 'c': ('3', '3')}
```

CSV files come in many different flavors. To define a new format with a different delimiter, string quoting convention, or line terminator, we define a simple subclass of `csv.Dialect`:

```
class my_dialect(csv.Dialect):
    lineterminator = '\n'
    delimiter = ';'
    quotechar = '"'
    quoting = csv.QUOTE_MINIMAL

reader = csv.reader(f, dialect=my_dialect)
```

We can also give individual CSV dialect parameters as keywords to `csv.reader` without having to define a subclass:

```
reader = csv.reader(f, delimiter='|')
```

텍스트 파일에서 데이터를 읽고 쓰는 법 > 구분자 형식 다루기

Table 6-3. CSV dialect options

Argument	Description
`delimiter`	One-character string to separate fields; defaults to `','`.
`lineterminator`	Line terminator for writing; defaults to `'\r\n'`. Reader ignores this and recognizes cross-platform line terminators.
`quotechar`	Quote character for fields with special characters (like a delimiter); default is `'"'`.
`quoting`	Quoting convention. Options include `csv.QUOTE_ALL` (quote all fields), `csv.QUOTE_MINIMAL` (only fields with special characters like the delimiter), `csv.QUOTE_NONNUMERIC`, and `csv.QUOTE_NONE` (no quoting). See Python's documentation for full details. Defaults to `QUOTE_MINIMAL`.
`skipinitialspace`	Ignore whitespace after each delimiter; default is `False`.
`doublequote`	How to handle quoting character inside a field; if `True`, it is doubled (see online documentation for full detail and behavior).
`escapechar`	String to escape the delimiter if `quoting` is set to `csv.QUOTE_NONE`; disabled by default.

For files with more complicated or fixed multicharacter delimiters, you will not be able to use the `csv` module. In those cases, you'll have to do the line splitting and other cleanup using string's `split` method or the regular expression method `re.split`.

To *write* delimited files manually, you can use `csv.writer`. It accepts an open, writable file object and the same dialect and format options as `csv.reader`:

```
with open('mydata.csv', 'w') as f:
    writer = csv.writer(f, dialect=my_dialect)
    writer.writerow(('one', 'two', 'three'))
    writer.writerow(('1', '2', '3'))
    writer.writerow(('4', '5', '6'))
    writer.writerow(('7', '8', '9'))
```

텍스트 파일에서 데이터를 읽고 쓰는 법 > JSON 데이터

JSON (short for JavaScript Object Notation) has become one of the standard formats for sending data by HTTP request between web browsers and other applications. It is a much more free-form data format than a tabular text form like CSV. Here is an example:

```
obj = """
{"name": "Wes",
 "places_lived": ["United States", "Spain", "Germany"],
 "pet": null,
 "siblings": [{"name": "Scott", "age": 30, "pets": ["Zeus", "Zuko"]},
              {"name": "Katie", "age": 38,
               "pets": ["Sixes", "Stache", "Cisco"]}]
}
"""
```

JSON is very nearly valid Python code with the exception of its null value `null` and some other nuances (such as disallowing trailing commas at the end of lists). The basic types are objects (dicts), arrays (lists), strings, numbers, booleans, and nulls. All of the keys in an object must be strings. There are several Python libraries for reading and writing JSON data. I'll use `json` here, as it is built into the Python standard library. To convert a JSON string to Python form, use `json.loads`:

```
In [62]: import json

In [63]: result = json.loads(obj)

In [64]: result
Out[64]:
{'name': 'Wes',
 'pet': None,
 'places_lived': ['United States', 'Spain', 'Germany'],
 'siblings': [{'age': 30, 'name': 'Scott', 'pets': ['Zeus', 'Zuko']},
  {'age': 38, 'name': 'Katie', 'pets': ['Sixes', 'Stache', 'Cisco']}]}
```

`json.dumps`, on the other hand, converts a Python object back to JSON:

```
In [65]: asjson = json.dumps(result)
```

How you convert a JSON object or list of objects to a DataFrame or some other data structure for analysis will be up to you. Conveniently, you can pass a list of dicts (which were previously JSON objects) to the DataFrame constructor and select a subset of the data fields:

```
In [66]: siblings = pd.DataFrame(result['siblings'], columns=['name', 'age'])

In [67]: siblings
Out[67]:
    name  age
0  Scott   30
1  Katie   38
```

텍스트 파일에서 데이터를 읽고 쓰는 법 > JSON 데이터

The `pandas.read_json` can automatically convert JSON datasets in specific arrangements into a Series or DataFrame. For example:

```
In [68]: !cat examples/example.json
[{"a": 1, "b": 2, "c": 3},
 {"a": 4, "b": 5, "c": 6},
 {"a": 7, "b": 8, "c": 9}]
```

The default options for `pandas.read_json` assume that each object in the JSON array is a row in the table:

```
In [69]: data = pd.read_json('examples/example.json')

In [70]: data
Out[70]:
   a  b  c
0  1  2  3
1  4  5  6
2  7  8  9
```

For an extended example of reading and manipulating JSON data (including nested records), see the USDA Food Database example in Chapter 7.

If you need to export data from pandas to JSON, one way is to use the `to_json` methods on Series and DataFrame:

```
In [71]: print(data.to_json())
{"a":{"0":1,"1":4,"2":7},"b":{"0":2,"1":5,"2":8},"c":{"0":3,"1":6,"2":9}}

In [72]: print(data.to_json(orient='records'))
[{"a":1,"b":2,"c":3},{"a":4,"b":5,"c":6},{"a":7,"b":8,"c":9}]
```

Python has many libraries for reading and writing data in the ubiquitous HTML and XML formats. Examples include lxml, Beautiful Soup, and html5lib. While lxml is comparatively much faster in general, the other libraries can better handle malformed HTML or XML files.

pandas has a built-in function, `read_html`, which uses libraries like lxml and Beautiful Soup to automatically parse tables out of HTML files as DataFrame objects. To show how this works, I downloaded an HTML file (used in the pandas documentation) from the United States FDIC government agency showing bank failures.[1] First, you must install some additional libraries used by `read_html`:

```
conda install lxml
pip install beautifulsoup4 html5lib
```

If you are not using conda, `pip install lxml` will likely also work.

If you are not using conda, `pip install lxml` will likely also work.

The `pandas.read_html` function has a number of options, but by default it searches for and attempts to parse all tabular data contained within `<table>` tags. The result is a list of DataFrame objects:

```
In [73]: tables = pd.read_html('examples/fdic_failed_bank_list.html')

In [74]: len(tables)
Out[74]: 1

In [75]: failures = tables[0]

In [76]: failures.head()
Out[76]:
                      Bank Name             City  ST   CERT  \
0                   Allied Bank          Mulberry  AR     91
1  The Woodbury Banking Company          Woodbury  GA  11297
2        First CornerStone Bank  King of Prussia  PA  35312
3             Trust Company Bank          Memphis  TN   9956
4    North Milwaukee State Bank        Milwaukee  WI  20364

                 Acquiring Institution        Closing Date       Updated Date
0                      Today's Bank  September 23, 2016  November 17, 2016
1                      United Bank      August 19, 2016  November 17, 2016
2  First-Citizens Bank & Trust Company         May 6, 2016  September 6, 2016
3          The Bank of Fayette County       April 29, 2016  September 6, 2016
4  First-Citizens Bank & Trust Company       March 11, 2016      June 16, 2016
```

Because `failures` has many columns, pandas inserts a line break character \.

As you will learn in later chapters, from here we could proceed to do some data cleaning and analysis, like computing the number of bank failures by year:

```
In [77]: close_timestamps = pd.to_datetime(failures['Closing Date'])

In [78]: close_timestamps.dt.year.value_counts()
Out[78]:
2010    157
2009    140
2011     92
2012     51
2008     25
       ...
2004      4
2001      4
2007      3
2003      3
2000      2
Name: Closing Date, Length: 15, dtype: int64
```

Parsing XML with lxml.objectify

XML (eXtensible Markup Language) is another common structured data format supporting hierarchical, nested data with metadata. The book you are currently reading was actually created from a series of large XML documents.

Earlier, I showed the `pandas.read_html` function, which uses either lxml or Beautiful Soup under the hood to parse data from HTML. XML and HTML are structurally similar, but XML is more general. Here, I will show an example of how to use lxml to parse data from a more general XML format.

The New York Metropolitan Transportation Authority (MTA) publishes a number of data series about its bus and train services. Here we'll look at the performance data, which is contained in a set of XML files. Each train or bus service has a different file (like *Performance_MNR.xml* for the Metro-North Railroad) containing monthly data as a series of XML records that look like this:

```
<INDICATOR>
  <INDICATOR_SEQ>373889</INDICATOR_SEQ>
  <PARENT_SEQ></PARENT_SEQ>
  <AGENCY_NAME>Metro-North Railroad</AGENCY_NAME>
  <INDICATOR_NAME>Escalator Availability</INDICATOR_NAME>
  <DESCRIPTION>Percent of the time that escalators are operational
  systemwide. The availability rate is based on physical observations performed
  the morning of regular business days only. This is a new indicator the agency
  began reporting in 2009.</DESCRIPTION>
  <PERIOD_YEAR>2011</PERIOD_YEAR>
  <PERIOD_MONTH>12</PERIOD_MONTH>
  <CATEGORY>Service Indicators</CATEGORY>
  <FREQUENCY>M</FREQUENCY>
  <DESIRED_CHANGE>U</DESIRED_CHANGE>
  <INDICATOR_UNIT>%</INDICATOR_UNIT>
  <DECIMAL_PLACES>1</DECIMAL_PLACES>
  <YTD_TARGET>97.00</YTD_TARGET>
  <YTD_ACTUAL></YTD_ACTUAL>
  <MONTHLY_TARGET>97.00</MONTHLY_TARGET>
  <MONTHLY_ACTUAL></MONTHLY_ACTUAL>
</INDICATOR>
```

Using `lxml.objectify`, we parse the file and get a reference to the root node of the XML file with `getroot`:

```
from lxml import objectify

path = 'examples/mta_perf/Performance_MNR.xml'
parsed = objectify.parse(open(path))
root = parsed.getroot()
```

`root.INDICATOR` returns a generator yielding each `<INDICATOR>` XML element. For each record, we can populate a dict of tag names (like `YTD_ACTUAL`) to data values (excluding a few tags):

```
data = []

skip_fields = ['PARENT_SEQ', 'INDICATOR_SEQ',
               'DESIRED_CHANGE', 'DECIMAL_PLACES']

for elt in root.INDICATOR:
    el_data = {}
    for child in elt.getchildren():
        if child.tag in skip_fields:
            continue
        el_data[child.tag] = child.pyval
    data.append(el_data)
```

Lastly, convert this list of dicts into a DataFrame:

```
In [81]: perf = pd.DataFrame(data)

In [82]: perf.head()
Out[82]:
Empty DataFrame
Columns: []
Index: []
```

XML data can get much more complicated than this example. Each tag can have metadata, too. Consider an HTML link tag, which is also valid XML:

```
from io import StringIO
tag = '<a href="http://www.google.com">Google</a>'
root = objectify.parse(StringIO(tag)).getroot()
```

You can now access any of the fields (like `href`) in the tag or the link text:

```
In [84]: root
Out[84]: <Element a at 0x7f6b15817748>

In [85]: root.get('href')
Out[85]: 'http://www.google.com'

In [86]: root.text
Out[86]: 'Google'
```

이진 데이터 형식

Binary Data Formats

One of the easiest ways to store data (also known as *serialization*) efficiently in binary format is using Python's built-in `pickle` serialization. pandas objects all have a `to_pickle` method that writes the data to disk in pickle format:

```
In [87]: frame = pd.read_csv('examples/ex1.csv')

In [88]: frame
Out[88]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo

In [89]: frame.to_pickle('examples/frame_pickle')
```

You can read any "pickled" object stored in a file by using the built-in `pickle` directly, or even more conveniently using `pandas.read_pickle`:

```
In [90]: pd.read_pickle('examples/frame_pickle')
Out[90]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

`pickle` is only recommended as a short-term storage format. The problem is that it is hard to guarantee that the format will be stable over time; an object pickled today may not unpickle with a later version of a library. We have tried to maintain backward compatibility when possible, but at some point in the future it may be necessary to "break" the pickle format.

pandas has built-in support for two more binary data formats: HDF5 and MessagePack. I will give some HDF5 examples in the next section, but I encourage you to explore different file formats to see how fast they are and how well they work for your analysis. Some other storage formats for pandas or NumPy data include:

bcolz
: A compressable column-oriented binary format based on the Blosc compression library.

Feather
: A cross-language column-oriented file format I designed with the R programming community's Hadley Wickham. Feather uses the Apache Arrow columnar memory format.

이진 데이터 형식 > HDF5 형식 사용하기

HDF5 is a well-regarded file format intended for storing large quantities of scientific array data. It is available as a C library, and it has interfaces available in many other languages, including Java, Julia, MATLAB, and Python. The "HDF" in HDF5 stands for *hierarchical data format*. Each HDF5 file can store multiple datasets and supporting metadata. Compared with simpler formats, HDF5 supports on-the-fly compression with a variety of compression modes, enabling data with repeated patterns to be stored more efficiently. HDF5 can be a good choice for working with very large datasets that don't fit into memory, as you can efficiently read and write small sections of much larger arrays.

While it's possible to directly access HDF5 files using either the PyTables or h5py libraries, pandas provides a high-level interface that simplifies storing Series and DataFrame object. The `HDFStore` class works like a dict and handles the low-level details:

```
In [92]: frame = pd.DataFrame({'a': np.random.randn(100)})

In [93]: store = pd.HDFStore('mydata.h5')

In [94]: store['obj1'] = frame

In [95]: store['obj1_col'] = frame['a']

In [96]: store
Out[96]:
<class 'pandas.io.pytables.HDFStore'>
File path: mydata.h5
/obj1                frame        (shape->[100,1])

/obj1_col            series       (shape->[100])

/obj2                frame_table  (typ->appendable,nrows->100,ncols->1,indexers->
[index])
/obj3                frame_table  (typ->appendable,nrows->100,ncols->1,indexers->
[index])
```

이진 데이터 형식 > HDF5 형식 사용하기

Objects contained in the HDF5 file can then be retrieved with the same dict-like API:

```
In [97]: store['obj1']
Out[97]:
          a
0  -0.204708
1   0.478943
2  -0.519439
3  -0.555730
4   1.965781
..       ...
95  0.795253
96  0.118110
97 -0.748532
98  0.584970
99  0.152677
[100 rows x 1 columns]
```

`HDFStore` supports two storage schemas, `'fixed'` and `'table'`. The latter is generally slower, but it supports query operations using a special syntax:

```
In [98]: store.put('obj2', frame, format='table')

In [99]: store.select('obj2', where=['index >= 10 and index <= 15'])
Out[99]:
          a
10  1.007189
11 -1.296221
12  0.274992
13  0.228913
14  1.352917
15  0.886429

In [100]: store.close()
```

The `put` is an explicit version of the `store['obj2'] = frame` method but allows us to set other options like the storage format.

The `pandas.read_hdf` function gives you a shortcut to these tools:

```
In [101]: frame.to_hdf('mydata.h5', 'obj3', format='table')
In [102]: pd.read_hdf('mydata.h5', 'obj3', where=['index < 5'])
Out[102]:
          a
0 -0.204708
1  0.478943
2 -0.519439
3 -0.555730
4  1.965781
```

If you are processing data that is stored on remote servers, like Amazon S3 or HDFS, using a different binary format designed for distributed storage like Apache Parquet may be more suitable. Python for Parquet and other such storage formats is still developing, so I do not write about them in this book.

If you work with large quantities of data locally, I would encourage you to explore PyTables and h5py to see how they can suit your needs. Since many data analysis problems are I/O-bound (rather than CPU-bound), using a tool like HDF5 can massively accelerate your applications.

HDF5 is *not* a database. It is best suited for write-once, read-many datasets. While data can be added to a file at any time, if multiple writers do so simultaneously, the file can become corrupted.

이진 데이터 형식 > 엑셀 파일에서 데이터 읽어오기

pandas also supports reading tabular data stored in Excel 2003 (and higher) files using either the `ExcelFile` class or `pandas.read_excel` function. Internally these tools use the add-on packages `xlrd` and `openpyxl` to read XLS and XLSX files, respectively. You may need to install these manually with pip or conda.

To use `ExcelFile`, create an instance by passing a path to an `xls` or `xlsx` file:

```
In [104]: xlsx = pd.ExcelFile('examples/ex1.xlsx')
```

Data stored in a sheet can then be read into DataFrame with `parse`:

```
In [105]: pd.read_excel(xlsx, 'Sheet1')
Out[105]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

If you are reading multiple sheets in a file, then it is faster to create the `ExcelFile`, but you can also simply pass the filename to `pandas.read_excel`:

```
In [106]: frame = pd.read_excel('examples/ex1.xlsx', 'Sheet1')

In [107]: frame
Out[107]:
   a   b   c   d message
0  1   2   3   4   hello
1  5   6   7   8   world
2  9  10  11  12     foo
```

To write pandas data to Excel format, you must first create an `ExcelWriter`, then write data to it using pandas objects' `to_excel` method:

```
In [108]: writer = pd.ExcelWriter('examples/ex2.xlsx')

In [109]: frame.to_excel(writer, 'Sheet1')

In [110]: writer.save()
```

You can also pass a file path to `to_excel` and avoid the `ExcelWriter`:

```
In [111]: frame.to_excel('examples/ex2.xlsx')
```

웹 API와 함께 사용하기

Interacting with Web APIs

Many websites have public APIs providing data feeds via JSON or some other format. There are a number of ways to access these APIs from Python; one easy-to-use method that I recommend is the `requests` package.

To find the last 30 GitHub issues for pandas on GitHub, we can make a GET HTTP request using the add-on `requests` library:

```
In [113]: import requests

In [114]: url = 'https://api.github.com/repos/pandas-dev/pandas/issues'

In [115]: resp = requests.get(url)

In [116]: resp
Out[116]: <Response [200]>
```

The Response object's `json` method will return a dictionary containing JSON parsed into native Python objects:

```
In [117]: data = resp.json()

In [118]: data[0]['title']
Out[118]: 'Period does not round down for frequencies less that 1 hour'
```

Each element in `data` is a dictionary containing all of the data found on a GitHub issue page (except for the comments). We can pass `data` directly to DataFrame and extract fields of interest:

```
In [119]: issues = pd.DataFrame(data, columns=['number', 'title',
   .....:                                      'labels', 'state'])

In [120]: issues
Out[120]:
    number                                              title  \
0    17666  Period does not round down for frequencies les...
1    17665           DOC: improve docstring of function where
2    17664               COMPAT: skip 32-bit test on int repr
3    17662                           implement Delegator class
4    17654  BUG: Fix series rename called with str alterin...
..     ...                                                ...
25   17603  BUG: Correctly localize naive datetime strings...
26   17599                     core.dtypes.generic --> cython
27   17596   Merge cdate_range functionality into bdate_range
28   17587  Time Grouper bug fix when applied for list gro...
29   17583  BUG: fix tz-aware DatetimeIndex + TimedeltaInd...

                                               labels state
0                                                  []  open
1   [{'id': 134699, 'url': 'https://api.github.com...  open
2   [{'id': 563047854, 'url': 'https://api.github....  open
3                                                  []  open
4   [{'id': 76811, 'url': 'https://api.github.com/...  open
..                                                ...   ...
25  [{'id': 76811, 'url': 'https://api.github.com/...  open
26  [{'id': 49094459, 'url': 'https://api.github.c...  open
27  [{'id': 35818298, 'url': 'https://api.github.c...  open
28  [{'id': 233160, 'url': 'https://api.github.com...  open
29  [{'id': 76811, 'url': 'https://api.github.com/...  open
[30 rows x 4 columns]
```

With a bit of elbow grease, you can create some higher-level interfaces to common web APIs that return DataFrame objects for easy analysis.

데이터베이스와 함께 사용하기

Interacting with Databases

In a business setting, most data may not be stored in text or Excel files. SQL-based relational databases (such as SQL Server, PostgreSQL, and MySQL) are in wide use, and many alternative databases have become quite popular. The choice of database is usually dependent on the performance, data integrity, and scalability needs of an application.

Loading data from SQL into a DataFrame is fairly straightforward, and pandas has some functions to simplify the process. As an example, I'll create a SQLite database using Python's built-in `sqlite3` driver:

```
In [121]: import sqlite3

In [122]: query = """
   .....: CREATE TABLE test
   .....: (a VARCHAR(20), b VARCHAR(20),
   .....:  c REAL,        d INTEGER
   .....: );"""

In [123]: con = sqlite3.connect('mydata.sqlite')

In [124]: con.execute(query)
Out[124]: <sqlite3.Cursor at 0x7f6b12a50f10>

In [125]: con.commit()
```

Then, insert a few rows of data:

```
In [126]: data = [('Atlanta', 'Georgia', 1.25, 6),
   .....:         ('Tallahassee', 'Florida', 2.6, 3),
   .....:         ('Sacramento', 'California', 1.7, 5)]

In [127]: stmt = "INSERT INTO test VALUES(?, ?, ?, ?)"

In [128]: con.executemany(stmt, data)
Out[128]: <sqlite3.Cursor at 0x7f6b15c66ce0>

In [129]: con.commit()
```

Most Python SQL drivers (PyODBC, psycopg2, MySQLdb, pymssql, etc.) return a list of tuples when selecting data from a table:

```
In [130]: cursor = con.execute('select * from test')

In [131]: rows = cursor.fetchall()

In [132]: rows
Out[132]:
[('Atlanta', 'Georgia', 1.25, 6),
 ('Tallahassee', 'Florida', 2.6, 3),
 ('Sacramento', 'California', 1.7, 5)]
```

데이터베이스와 함께 사용하기

Interacting with Databases

You can pass the list of tuples to the DataFrame constructor, but you also need the column names, contained in the cursor's `description` attribute:

```
In [133]: cursor.description
Out[133]:
(('a', None, None, None, None, None, None),
 ('b', None, None, None, None, None, None),
 ('c', None, None, None, None, None, None),
 ('d', None, None, None, None, None, None))

In [134]: pd.DataFrame(rows, columns=[x[0] for x in cursor.description])
Out[134]:
             a           b     c  d
0      Atlanta     Georgia  1.25  6
1  Tallahassee     Florida  2.60  3
2   Sacramento  California  1.70  5
```

This is quite a bit of munging that you'd rather not repeat each time you query the database. The SQLAlchemy project is a popular Python SQL toolkit that abstracts away many of the common differences between SQL databases. pandas has a `read_sql` function that enables you to read data easily from a general SQLAlchemy connection. Here, we'll connect to the same SQLite database with SQLAlchemy and read data from the table created before:

```
In [135]: import sqlalchemy as sqla

In [136]: db = sqla.create_engine('sqlite:///mydata.sqlite')

In [137]: pd.read_sql('select * from test', db)
Out[137]:
             a           b     c  d
0      Atlanta     Georgia  1.25  6
1  Tallahassee     Florida  2.60  3
2   Sacramento  California  1.70  5
```

Q&A and Break Time

질의응답 및 휴식 시간 (5분)

2부: 파일 형식 & DB 등 최적화

1) Columnar File Format

2) Column oriented DB

3) 메모리 & 디스크 관리법

노하우 리스트업

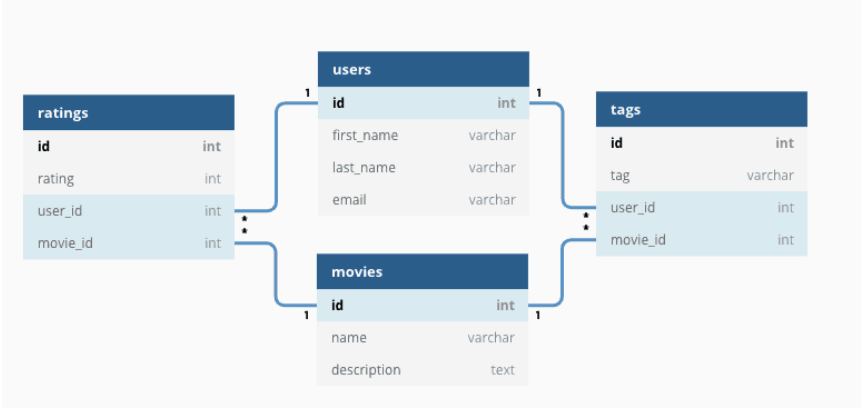
Know-How List

- MLOps
- AutoML Tool / Case Study
- Data Analytics Tool
- Artificial Intelligence History
- **Memory Optimization**
- **Disk Optimization**
- **SQL/NoSQL Database Ranking**
- 개발환경 구축 방법론 Case Study
- **인공지능(Data&ML&DL) 직업군 분류 / 연봉 수준 / 요구 수준**
- **취업 관련 (스타트업 & 대기업 / SmartCity / Fintech / AI)**

SQL

Structured Query Language

- 관계형 데이터베이스 관리 시스템(RDBMS)의 데이터를 관리하기 위해 설계된 특수 목적의 프로그래밍 언어
  - 자료의 검색과 관리
  - 데이터베이스 스키마 생성과 수정
  - 데이터베이스 객체 접근 조정 관리
- 많은 수의 데이터베이스 관련 프로그램들이 SQL을 표준으로 채택하고 있다.



DB 종류

Kinds of Database

- Relational DBMS
- Key-value stores
- Document stores
- Time Series DBMS
- Graph DBMS
- Object oriented DBMS
- Search engines
- RDF stores
- Wide column stores
- Multivalue DBMS
- Native XML DBMS
- Spatial DBMS
- Event Stores
- Content stores
- Navigational DBMS

SQL 구문

Structured Query Language

- 데이터 정의 언어 (DDL : Data Definition Language)
- 데이터 조작 언어 (DML : Data Manipulation Language)
- 데이터 제어 언어 (DCL : Data Control Language)

SQL 구문

Structured Query Language

데이터 정의 언어 (DDL : Data Definition Language)

- CREATE (데이터베이스 개체 (테이블, 인덱스, 제약조건 등)의 정의)
- DROP (데이터베이스 개체 삭제)
- ALTER (데이터베이스 개체 정의 변경)

```sql
CREATE TABLE My_table(
 my_field1 INT,
 my_field2 VARCHAR(50),
 my_field3 DATE NOT NULL,
 PRIMARY KEY (my_field1, my_field2)
);
```

- `ALTER` 는 다양한 방법으로 현존하는 객체의 구조를 변경한다. 예를 들어, 현재의 테이블에 컬럼을 추가하거나 제한을 추가한다. 예를 들면:

```sql
ALTER TABLE My_table ADD my_field4 NUMBER(3) NOT NULL;
```

- `TRUNCATE` 는 테이블에서 모든 데이터를 빠르게 삭제한다. 내부 테이블의 데이터를 삭제하는 것이지, 테이블 자체를 삭제하는 것은 아니다. 이 과정은 보통 순차적인 COMMIT 연산을 내포한다. 예를 들어, 이것은 롤백되지 않는다. (DELETE와는 달리 이후 롤백을 위한 로그가 기록되지 않는다.).

```sql
TRUNCATE TABLE My_table;
```

- `DROP` 은 데이터베이스에서 객체를 삭제하며, 보통 롤백을 통해 복원할 수 없다. 예를 들어:

```sql
DROP TABLE My_table;
```

SQL 구문

Structured Query Language

데이터 조작 언어 (DML : Data Manipulation Language)

- INSERT INTO (행 데이터 또는 테이블 데이터의 삽입)
- UPDATE ~ SET (표 업데이트)
- DELETE FROM (테이블에서 특정 행 삭제)
- SELECT ~ FROM ~ WHERE (테이블 데이터의 검색 결과 집합의 취득)

SQL 구문

Structured Query Language

데이터 제어 언어 (DCL : Data Control Language)

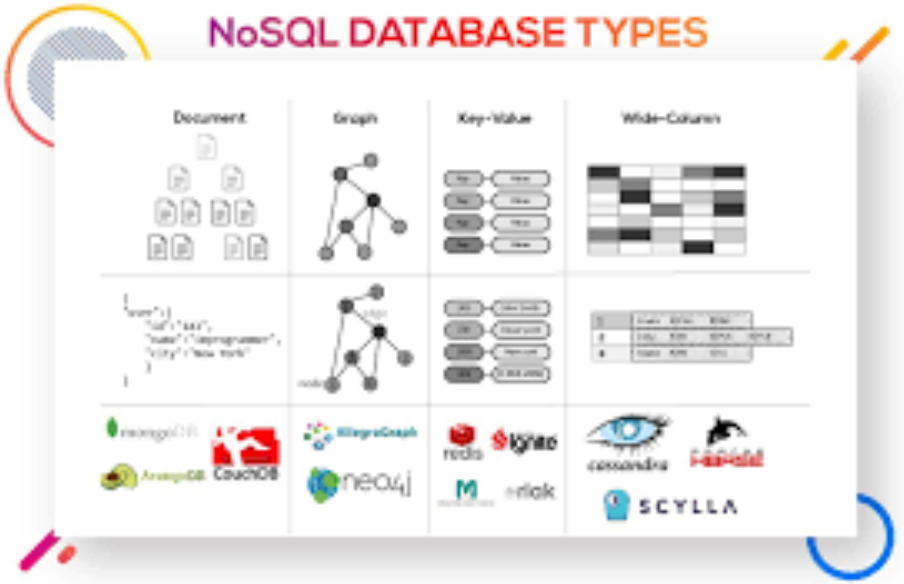
- GRANT (특정 데이터베이스 사용자에게 특정 작업을 수행 권한을 부여)
- REVOKE (특정 데이터베이스 이용자로부터 이미 준 권한을 박탈 함.)
- SET TRANSACTION (트랜잭션 모드 설정 (동시 트랜잭션 격리 수준 (ISOLATION MODE) 등))
- BEGIN (트랜잭션 시작)
- COMMIT (트랜잭션의 실행)
- ROLLBACK (트랜잭션 취소)
- SAVEPOINT (무작위로 롤백 지점을 설정)
- LOCK (TABLE 등의 자원을 차지)

SQL VS NoSQL

Structured Query Language VS Non Structured Query Language

NoSQL

- 전통적인 관계형 데이터베이스 보다 덜 제한적인 일관성 모델을 이용하는 데이터의 저장 및 검색을 위한 매커니즘을 제공
- 디자인의 단순화, 수평적 확장성, 세세한 통제를 포함
- 단순 검색 및 추가 작업을 위한 매우 최적화된 키 값 저장 공간으로, 레이턴시와 스루풋과 관련하여 상당한 성능 이익을 내는 것이 목적
- 빅데이터와 실시간 웹 애플리케이션의 상업적 이용에 널리 쓰임

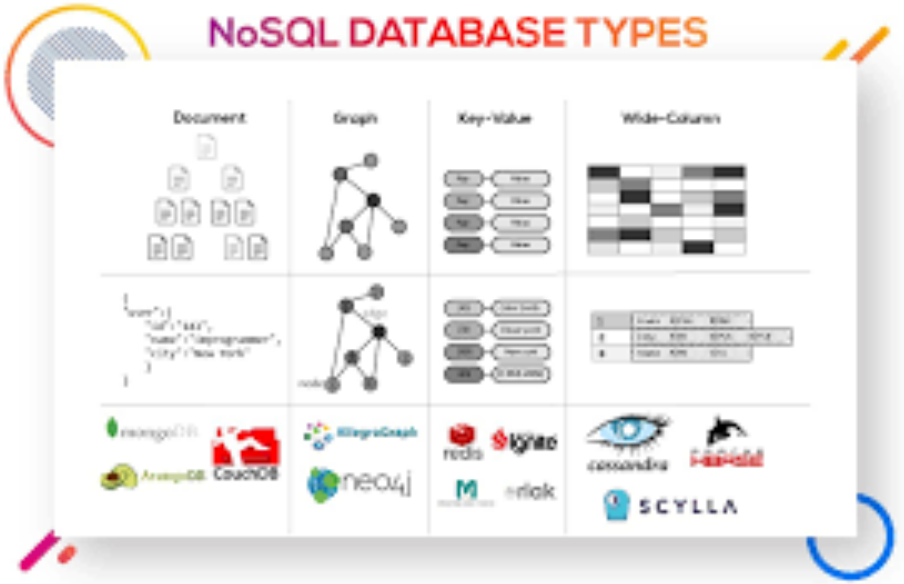


SQL VS NoSQL

Structured Query Language VS Non Structured Query Language

NoSQL의 종류

- 와이드 컬럼 스토어: H베이스, 아큐물로, 카산드라
- 도큐먼트: 몽고DB, 카우치베이스
- 키 값: 다이나모DB[5], 리악, 레디스, 캐시, 프로젝트 볼드모트
- 그래프: Neo4J, AgensGraph, 알레그로그래프, 버투오소



SQL VS NoSQL

Structured Query Language VS Non Structured Query Language

NoSQL 성능등급표 by 벤 스코필드

벤 스코필드는 여러 유형의 NoSQL 데이터베이스의 등급을 다음과 같이 평가했다:[17]

데이터 모델	성능	확장성	유연성	복잡성	기능
키-값 스토어	높음	높음	높음	없음	가변적 (없음)
컬럼 지향 스토어	높음	높음	준수	낮음	최소
도큐먼트 지향 스토어	높음	가변적 (높음)	높음	낮음	가변적 (낮음)
그래프 데이터베이스	가변적	가변적	높음	높음	그래프 이론
관계형 데이터베이스	가변적	가변적	낮음	준수	관계대수

SQL VS NoSQL

Structured Query Language VS Non Structured Query Language

Database Engine Ranking 2021년 8월 기준

373 systems in ranking, August 2021

Rank			DBMS	Database Model	Score		
Aug 2021	Jul 2021	Aug 2020			Aug 2021	Jul 2021	Aug 2020
1.	1.	1.	Oracle	Relational, Multi-model	1269.26	+6.59	-85.90
2.	2.	2.	MySQL	Relational, Multi-model	1238.22	+9.84	-23.36
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	973.35	-8.61	-102.53
4.	4.	4.	PostgreSQL	Relational, Multi-model	577.05	-0.10	+40.28
5.	5.	5.	MongoDB	Document, Multi-model	496.54	+0.38	+52.98
6.	6.	↑ 7.	Redis	Key-value, Multi-model	169.88	+1.58	+17.01
7.	7.	↓ 6.	IBM Db2	Relational, Multi-model	165.46	+0.31	+3.01
8.	8.	8.	Elasticsearch	Search engine, Multi-model	157.08	+1.32	+4.76
9.	9.	9.	SQLite	Relational	129.81	-0.39	+3.00
10.	↑ 11.	10.	Microsoft Access	Relational	114.84	+1.39	-5.02
11.	↓ 10.	11.	Cassandra	Wide column	113.66	-0.35	-6.18
12.	12.	12.	MariaDB	Relational, Multi-model	98.98	+0.99	+8.06
13.	13.	13.	Splunk	Search engine	90.60	+0.55	+0.69
14.	14.	↑ 15.	Hive	Relational	83.93	+1.26	+8.64
15.	15.	↑ 17.	Microsoft Azure SQL Database	Relational, Multi-model	75.15	-0.06	+18.31
16.	16.	16.	Amazon DynamoDB	Multi-model	74.90	-0.30	+10.15
17.	17.	↓ 14.	Teradata	Relational, Multi-model	68.82	-0.13	-7.96
18.	18.	↑ 21.	Neo4j	Graph	56.95	-0.21	+6.77
19.	19.	19.	SAP HANA	Relational, Multi-model	55.57	+1.76	+2.46
20.	20.	20.	Solr	Search engine, Multi-model	51.06	-0.73	-0.63

db ranking link

SQL VS NoSQL

Structured Query Language VS Non Structured Query Language

Database Engine Ranking 2021년 8월 기준

415 systems in ranking, October 2023

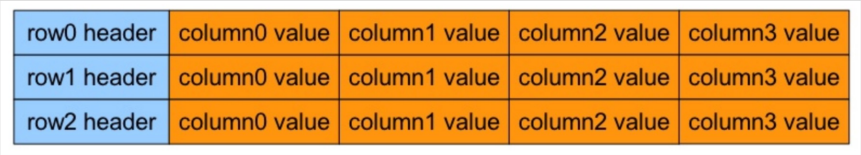
Rank			DBMS	Database Model	Score		
Oct 2023	Sep 2023	Oct 2022			Oct 2023	Sep 2023	Oct 2022
1.	1.	1.	Oracle	Relational, Multi-model	1261.42	+20.54	+25.05
2.	2.	2.	MySQL	Relational, Multi-model	1133.32	+21.83	-72.06
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	896.88	-5.34	-27.80
4.	4.	4.	PostgreSQL	Relational, Multi-model	638.82	+18.06	+16.10
5.	5.	5.	MongoDB	Document, Multi-model	431.42	-8.00	-54.81
6.	6.	6.	Redis	Key-value, Multi-model	162.96	-0.72	-20.41
7.	7.	7.	Elasticsearch	Search engine, Multi-model	137.15	-1.84	-13.92
8.	8.	8.	IBM Db2	Relational, Multi-model	134.87	-1.85	-14.79
9.	9.	↑ 10.	SQLite	Relational	125.14	-4.06	-12.66
10.	10.	↓ 9.	Microsoft Access	Relational	124.31	-4.25	-13.85
11.	11.	↑ 13.	Snowflake	Relational	123.24	+2.35	+16.51
12.	12.	↓ 11.	Cassandra	Wide column, Multi-model	108.82	-1.24	-9.12
13.	13.	↓ 12.	MariaDB	Relational, Multi-model	99.66	-0.79	-9.65
14.	14.	14.	Splunk	Search engine	92.37	+0.98	-2.28
15.	15.	↑ 16.	Microsoft Azure SQL Database	Relational, Multi-model	80.93	-1.80	-4.03
16.	16.	↓ 15.	Amazon DynamoDB	Multi-model	80.91	+0.00	-7.44
17.	17.	↑ 20.	Databricks	Multi-model	75.82	+0.64	+18.21
18.	18.	↓ 17.	Hive	Relational	69.18	-2.65	-11.42
19.	19.	↓ 18.	Teradata	Relational, Multi-model	58.56	-1.77	-7.51
20.	20.	↑ 22.	Google BigQuery	Relational	56.57	+0.11	+4.12
21.	21.	↑ 23.	FileMaker	Relational	53.32	-0.29	+0.91
22.	22.	↑ 24.	SAP HANA	Relational, Multi-model	49.44	-1.17	-2.63
23.	23.	↓ 19.	Neo4j	Graph	48.44	-1.95	-10.25
24.	24.	↓ 21.	Solr	Search engine, Multi-model	45.36	-1.73	-8.14
25.	25.	25.	SAP Adaptive Server	Relational, Multi-model	41.92	-1.40	-1.05

db ranking link

Row Oriented DB VS Column Oriented DB

Row Oriented DB

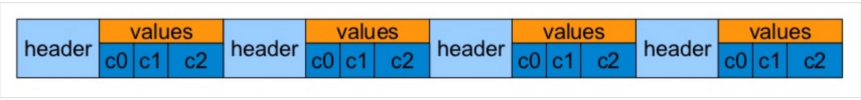
Row-Oriented DBMS는 데이터를 어떻게 저장할까?



- 주로 Row-Value 데이터를 연속적으로 저장한다.

실제로 테이블의 데이터파일 모습은?

- 일렬로 저장된 모습을 확인할 수 있다.



- Row-Value의 오프셋은 Value의 자료구조에 의존적이다.
  - `32bit의 Integer`, `64bit의 Integer`, `String` 등 자료구조의 크기의 총 합산 크기에 따라 오프셋을 결정한다.

Row Oriented DB 장단점

Row-Oriented를 정리하면

- 장점
  - 하나의 Row 데이터를 추가/삭제할때 하나의 페이지만 사용하기에 좋다.
  - 하나의 Row의 대부분의 컬럼을 읽어야 하거나 변경하는 경우 하나의 Read/Write로 가능하다.
- 단점
  - 모든 Row를 읽는 경우 불필요한 Column값을 다 읽어야 한다.
  - Page의 사용하지 않는 공간까지 읽어야 한다.

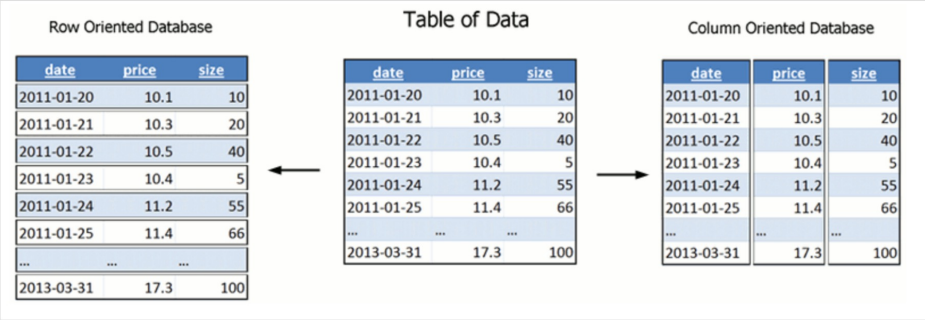
Row Oriented DB VS Column Oriented DB

Column Oriented DB

Column Oriented DBMS란?

- `Row` 형으로 저장하는 대신 `Column` 으로 저장하는 방식이다.
- 모든 `Column` 들은 개별적으로 다뤄지며 `Column` 별로 연속적으로 저장된다.
- `Column` 별로 데이터가 저장되기때문에 압축에도 높은 효율을 얻을 수 있다.

Row Oriented vs Column Oriented



date	price	size
2011-01-20	10.1	10
2011-01-21	10.3	20
2011-01-22	10.5	40
2011-01-23	10.4	5
2011-01-24	11.2	55
2011-01-25	11.4	66
...	...	...
2013-03-31	17.3	100

- 우리가 평균 가격을 알고 싶다면?
  - `Row Oriented` 는 모든 Row를 조회해야만 결과를 얻을 수 있다.
  - 하지만 `Column Oriented` 는 Price 컬럼만 조회해도 결과를 얻을 수 있다.

Column Oriented DB 장단점

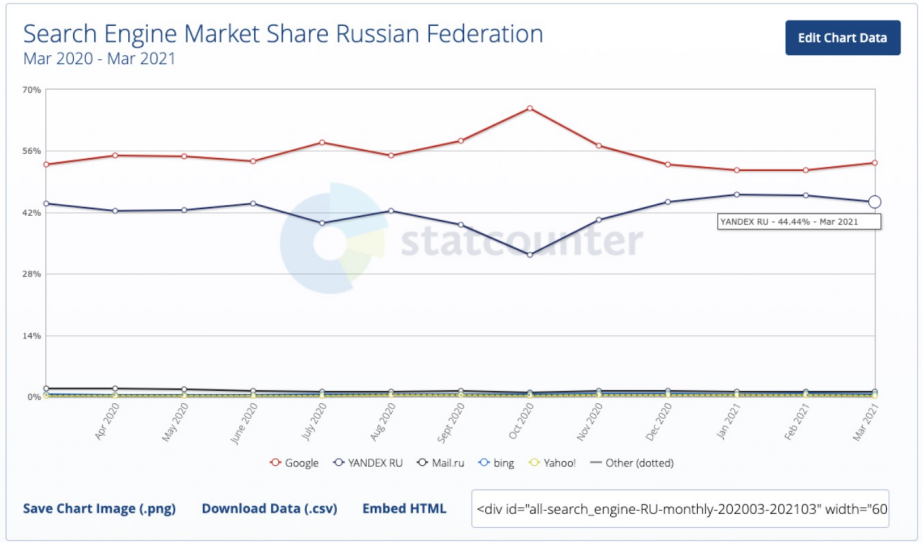
항상 Column Oriented가 좋을까?

- 다수의 컬럼을 조회하는 상황이라면 쓸모가 없어보인다.
- 오히려 결과를 합치는 작업 비용이 더 커져서 느릴 경우가 발생할 거 같다.

Column Oriented DB의 대표: Clickhouse DB

Clickhouse DB : Yandex가 자체 개발한 DB

(국내에서 러시아의 Naver라 불리우는 회사, 러시아 서치엔진 점유율 2위, 점유율 참조 링크)



- ClickHouse®는 온라인 쿼리 분석 처리(OLAP:online analytical processing of queries)를 위한 열 지향 (column-oriented) 데이터베이스 관리 시스템 (DBMS)
  (https://clickhouse.tech/docs/en/ 참조)
- Yandex Metrica, 현재, 글로벌 3위 웹 분석 플랫폼의 초기 개발을 위해서 만들어진 DB
  (https://clickhouse.tech/docs/en/introduction/history/ 참조)
- Column-oriented는 OLAP 시나리오에 더 적합하며, 대부분의 쿼리를 처리 할 때 최소 100 배 더 빠르다.

장점	단점
진정한 컬럼 지향 데이터베이스 관리 시스템	transaction에서 충분히 자격을 갖추진 못함
데이터 압축	빠른 속도와 짧은 지연 시간으로 이미 삽입 된 데이터를 수정하거나 삭제할 수 있는 능력 부재 * GDPR 준수를 위해 데이터 정리 혹은 수정시 사용할 수 있는 일괄 삭제 및 업데이트 기능이 있음
데이터의 디스크 스토리지	
다중 코어에서 병렬 처리	희소 색인은 ClickHouse가 Key로 단일 행을 검색하는 포인트 쿼리에 효율적이지 않음
여러 서버의 분산 처리	
SQL 지원	
벡터 계산 엔진	
실시간 데이터 업데이트	
기본 색인	
보조 인덱스	
온라인 쿼리에 적합	
근사 계산 지원	
적응 형 조인 알고리즘	
데이터 복제 및 데이터 무결성 지원	
역할 기반 액세스 제어	

Column Oriented DB의 대표: Clickhouse DB

Clickhouse 속도

https://clickhouse.tech/benchmark/dbms/

Column Oriented DB의 대표: Clickhouse DB

Clickhouse 인지도

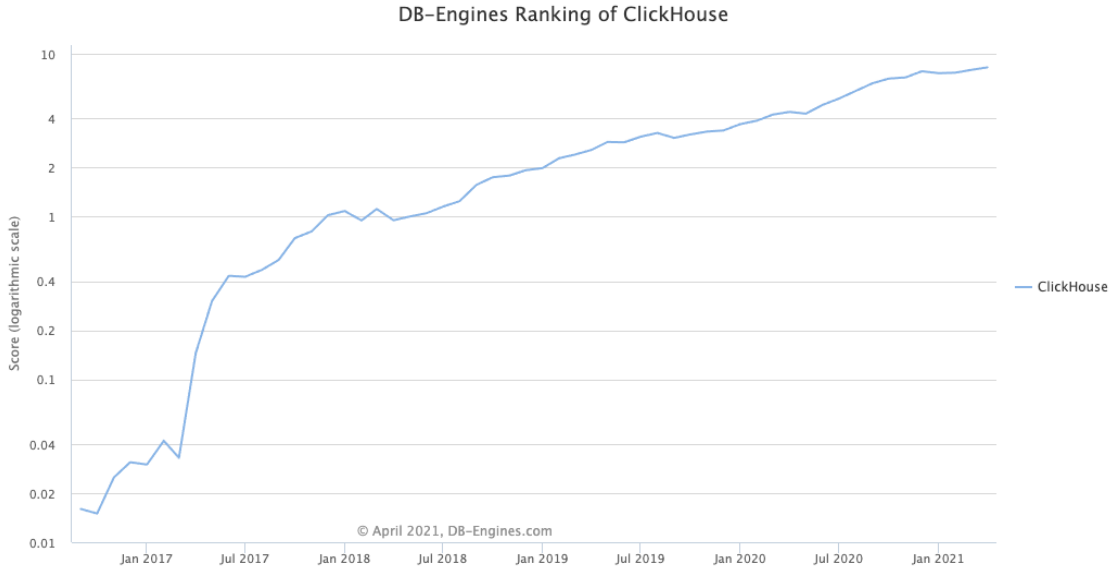
Ranking ClickHouse > Trend

DB-Engines Ranking - Trend of ClickHouse Popularity

The DB-Engines Ranking ranks database management systems according to their popularity.

This is a partial trend diagram of the complete ranking showing only ClickHouse.

Read more about the method of calculating the scores.



Ranking > Trend

RSS RSS Feed

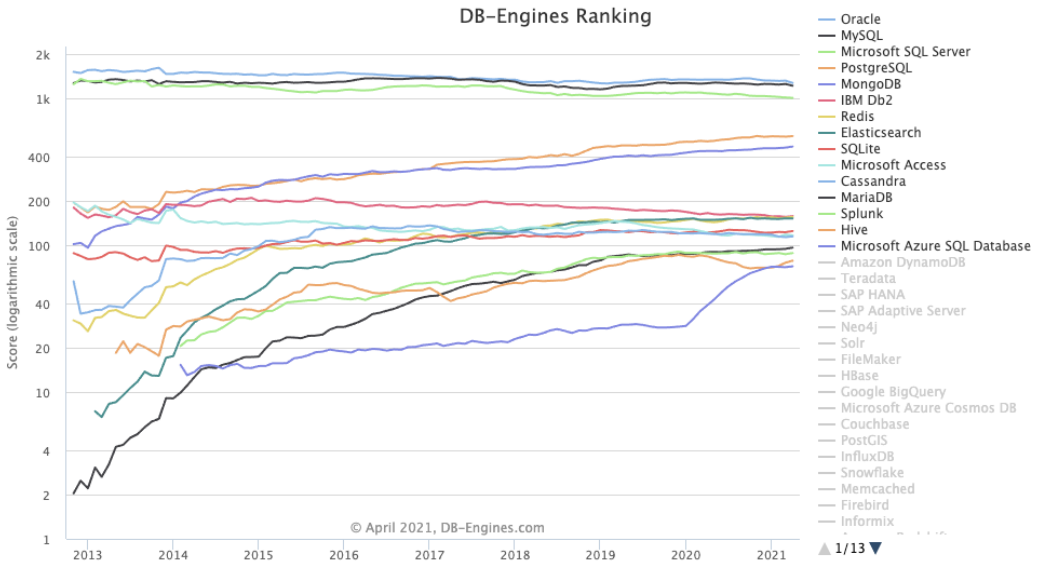
DB-Engines Ranking - Trend Popularity

The DB-Engines Ranking ranks database management systems according to their popularity.

Read more about the method of calculating the scores.

Rank	Trend	System	Score	Change
1		Oracle	1560	+ 27
2		MySQL	1342	+ 47
3		SQL Server	1278	- 40
4		PostgreSQL	174	- 3
5		MS Access	161	- 8
6		DB2	155	- 4

ranking table
April 2021



Click at a system in the legend
to hide or show its trend line

Column Oriented DB의 대표: Clickhouse DB

Clickhouse 인지도

ClickhouseDB는 현재 Tesla, Uber, Tencent, Kakao Corp, Alibaba cloud, eBay, Bloomberg에서도 활용중으로 알려져있음

kakaocorp	Internet company	—	—	—	if(kakao)2020 conference
ООО «МПЗ Богородский»	Agriculture	—	—	—	Article in Russian, November 2020
Tesla	Electric vehicle and clean energy company	—	—	—	Vacancy description, March 2021

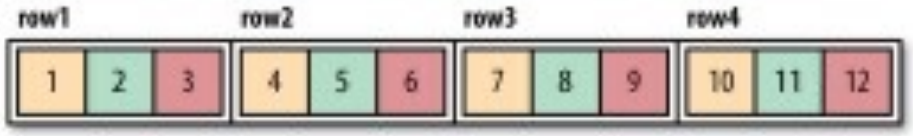
Row Oriented File VS Column Oriented File

비교 기준

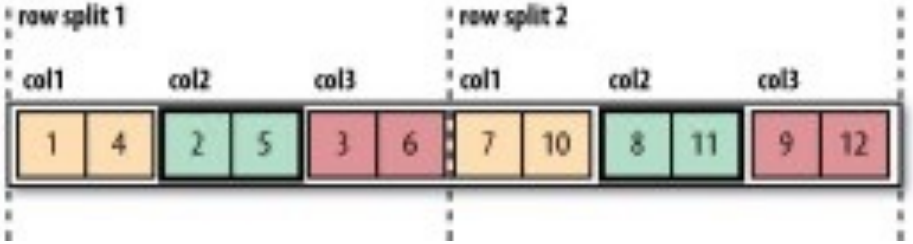
Logical table

	col1	col2	col3
row1	1	2	3
row2	4	5	6
row3	7	8	9
row4	10	11	12

Row-oriented layout (SequenceFile)



Column-oriented layout (RCFile)



Row Oriented File VS Column Oriented File

유명한 File Format 과 Parquet 비교

Spark Format Showdown		File Format		
		CSV	JSON	Parquet
Attribute	Columnar	No	No	Yes
	Compressable	Yes	Yes	Yes
	Splittable	Yes*	Yes**	Yes
	Human Readable	Yes	Yes	No
	Nestable	No	Yes	Yes
	Complex Data Structures	No	Yes	Yes
	Default Schema: Named columns	Manual	Automatic (full read)	Automatic (instant)
	Default Schema: Data Types	Manual (full read)	Automatic (full read)	Automatic (instant)

Row Oriented File VS Column Oriented File

Bigdata File Format 비교

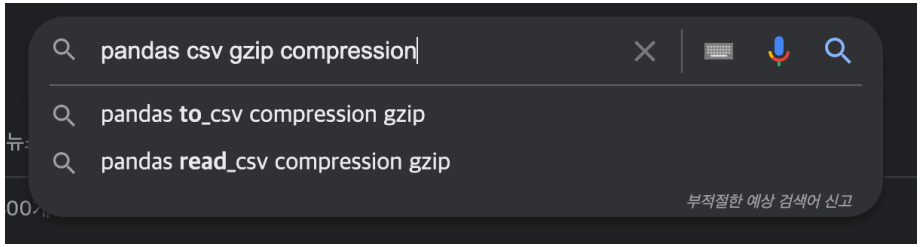
BIG DATA FORMATS COMPARISON

	Avro	Parquet	ORC
Schema Evolution Support			
Compression			
Splitability			
Most Compatible Platforms	Kafka, Druid	Impala, Arrow Drill, Spark	Hive, Presto
Row or Column	Row	Column	Column
Read or Write	Write	Read	Read

Source: Nexla analysis, April 2018

Memory / Disk Optimization

- Dataframe 메모리 최적화
  - Data Type 인식 및 메모리 최소화 진행
- CSV을 GZIP으로 압축하기
  - CSV를 꼭 사용해야한다면, 압축해서 저장 및 로딩
- Parquet 활용하기
  - 속도 최소 / 용량 최소 / 메모리 최소
  - DataFrame 메모리 최적화와 함께하면 최고의 성능 확인 가능
- 소스코드로 노하우 추가 공유




```
preprocessor.dataframe_cleansed.to_parquet(
    mlpath.df_cleansed_parquet, index=False
)
```

```
preprocessor.dataframe_cleansed = pandas.read_parquet(mlpath.df_cleansed_parquet)
```

[구글검색] pandas read chunksize

Optimize Pandas Memory Usage for Large Datasets

[github] csv_to_parquet.py

Q&A

질의응답 (5분)

끝. 감사합니다.

수업 듣느라 수고하셨습니다.